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INDIAN INSTITUTE OF TECHNOLOGY
KHARAGPUR

Stamp / Signature of the Invigilator	

EXAMINATION (End Semester)									SEMESTER (Spring)			
Roll Number									Section		Name	
Subject Number	C	S	3	0	2	0	2	Subject Name	*Database Management Systems*			
Department / Center of the Student									Additional sheets			

Important Instructions and Guidelines for Students

1. You must occupy your seat as per the Examination Schedule/Sitting Plan.
2. Do not keep mobile phones or any similar electronic gadgets with you even in the switched off mode.
3. Loose papers, class notes, books or any such materials must not be in your possession, even if they are irrelevant to the subject you are taking examination.
4. Data book, codes, graph papers, relevant standard tables/charts or any other materials are allowed only when instructed by the paper-setter.
5. Use of instrument box, pencil box and non-programmable calculator is allowed during the examination. However, exchange of these items or any other papers (including question papers) is not permitted.
6. Write on both sides of the answer script and do not tear off any page. **Use last page(s) of the answer script for rough work.** Report to the invigilator if the answer script has torn or distorted page(s).
7. It is your responsibility to ensure that you have signed the Attendance Sheet. Keep your Admit Card/Identity Card on the desk for checking by the invigilator.
8. You may leave the examination hall for wash room or for drinking water for a very short period. Record your absence from the Examination Hall in the register provided. Smoking and the consumption of any kind of beverages are strictly prohibited inside the Examination Hall.
9. Do not leave the Examination Hall without submitting your answer script to the invigilator. **In any case, you are not allowed to take away the answer script with you.** After the completion of the examination, do not leave the seat until the invigilators collect all the answer scripts.
10. During the examination, either inside or outside the Examination Hall, gathering information from any kind of sources or exchanging information with others or any such attempt will be treated as **'unfair means'**. Do not adopt unfair means and do not indulge in unseemly behavior.

Violation of any of the above instructions may lead to severe punishment.

Signature of the Student

To be filled in by the examiner

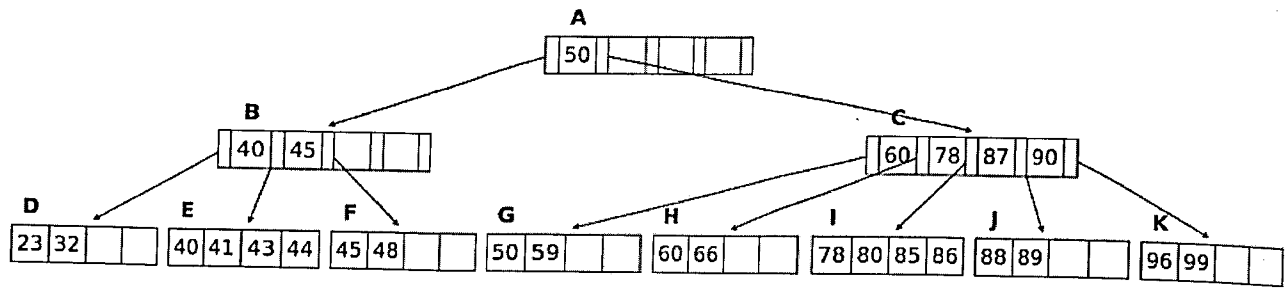
Question Number	1	2	3	4	5	6	7	8	9	10	Total
Marks Obtained											

Marks obtained (in words)	Signature of the Examiner	Signature of the Scrutineer

1.(a). Illustrate the page structure used for storing variable length records in a file. [5]

(b). A database uses B+ tree indexing. The disk has block size of 4KB. The size of the search key is 12 bytes and pointer (node/disk) is 8 bytes. The database has 1,000,000 records. Assume that each record fits into one disk block. What is the minimum number of disk block accesses required to retrieve any record in the database? Explain your answer. [5]

(c) Consider the following B+ tree, with nodes named A-K. Suppose a new data with key value 79 is to be inserted into the database. Draw the new B+ tree and mark the nodes that are updated.



2. Consider the join $R \bowtie_{R.a=S.b} S$ and the following information about the relations R and S to be joined. Relation R contains 10,000 tuples and has 10 tuples per block. Relation S contains 2000 tuples and also has 10 tuples per block. Attribute b of relation S is the primary key for S. Both relations are stored in simple heap files. Neither relation has any index built on it. 52 buffer blocks are available. Cost metric considered is the number of block I/Os. The cost of writing out the result is uniformly ignored. [4 X 5]

(a) What is the cost of joining R and S using a simple nested loop join? Explain.

(b) What is the cost using a block-nested loops join? Explain.

(c) What is the cost using a sort-merge join? Explain.

(d) How many tuples does the join of *R* and *S* produce, at most, and how many blocks are required to store the result of the join back on disk.

(e) Would your answer to any of the previous questions change if you were told that *R.a* is a foreign key that refers to *S.b*? Explain.

3.(a). State three rules for query transformation that may be used for query optimization. [5]

(b). Consider the following schema: `Sailors(sid, sname, rating, age); Reserves(sid, did, day);Boats(bid, bname, size)`. Also, `Reserves.sid` is a foreign key to `Sailors` and `Reserves.bid` is a foreign key to `Boats.bid`.

We are given the following information about the database: `Reserves` contains 10,000 records with 40 records per page. `Sailors` contains 1000 records with 20 records per page. `Boats` contains 100 records with 10 records per page. There are 50 values for `Reserves.bid`. There are 10 values for `Sailors.rating` (1...10). There are 10 values for `Boat.size`. There are 500 values for `Reserves.day`. Consider the following query:

```
SELECT S.sid, S.sname, B.bname  FROM Sailors S, Reserves R, Boats B
WHERE S.sid = R.sid AND R.bid = B.bid AND
B.size > 5 AND R.day = 'January 26, 2023';
```

Assuming uniform distribution of values and column independence, estimate the number of tuples returned by this query. Explain your answer. [5]

(c). Consider the relation *R*(A, B, C, D). The following set of functional dependencies hold: AB → C, BC → D, CD → A, and AD → B. What are the non-trivial functional dependencies that follow from these FDs? What are the superkeys of *R*? You need to write only the final answer. [10]

4.(a). Clearly state the conditions for conflict equivalence between a pair of schedules. [5]

(b). Two transactions T_1 and T_2 consists of the following operations:

T_1: $r_1(A)$, $w_1(A)$, $r_1(B)$, $w_1(B)$; T_2: $r_2(B)$, $w_2(B)$, $r_2(C)$, $w_2(C)$;

Here, $r_i(X)$ and $w_i(X)$ denotes read and write by transaction T_i on data item X. What is the total number of conflict serializable schedules that are possible considering T_1 and T_2? Explain. [5]

(c). Consider the following schedule involving three transactions T_1, T_2, and T_3:

Start, $r_1(X)$, $w_2(X)$, $w_1(X)$, $r_3(X)$, T_1:*Commit*, T_2:*Commit*, T_3:*Commit*

State with explanation, which of the following properties hold for the schedule: [5 X 2]

Property	*Yes/No*	*Explanation*
Conflict serializable		

Property	*Yes/No*	*Explanation*
Recoverable		
Avoid cascade rollback		
Allowed by 2PL		If allowed write the schedule with locking commands
Allowed by Strict 2PL		If allowed write the schedule with locking commands

5.(a). Explain the wait-die scheme for deadlock prevention. [5]

(b). Consider the log records written by transactions T1 and T2: <START T1>; <T1, A, 10, 11>; <START T2>; <T2, B, 20, 21>; <T1, C, 30, 31>; <T2, D, 40, 41>; <COMMIT, T2>; <T1, E, 50, 51>; <COMMIT, T1>

What are the changes made by the recovery manager to the disk and the log during recovery from crash, if the last log record to appear on disk is <T1, E, 50, 51>. [5]

Database Changes:
Log Record Changes:

(c). Consider a timestamp based concurrency controller. Below are several sequences of events, including start events, where *sti* means that transaction *Ti* starts and *coi* means *Ti* commits. These sequences represent real time, and the timestamp-based scheduler will allocate timestamps to transactions in the order of their starts. In each case below, say what happens with the last request. You have to choose between one of the following three possible outcomes: the request is 1. accepted, 2. ignored, 3. transaction is rolled back. [10]

Sequence: *st1; st2; st3; r1(A); w1(A); r2(A);*	
Outcome:	Explanation:
Sequence: *st1; st2; r2(A); co2; r1(A); w1(A);*	
Outcome:	Explanation:
Sequence: *st1; st2; st3; r1(A); w2(A); w3(A); r2(A);*	
Outcome:	Explanation:
Sequence: *st1; st2; r1(A); r2(A); w1(B); w2(B);*	
Outcome:	Explanation:
Sequence: *st1; st2; st3; r1(A); w3(A); co3; r2(B); w2(A);*	
Outcome:	Explanation: